Indian Institute of Management
Lucknow

JSW Challenge 2023

National Winner Case Deck

TECHNOLOGY STRATEGY FOR
SUSTAINABLE STEEL

Driving a sustainable future: Harnessing enablers to combat the emissions challenge across entire value chain, with real-time monitoring and carbon offset via our integrated dashboard

ISSUES

CO2 emission (per ton of steel)

2.55 tonnes (INDIA)

1.85 tonnes (GLOBAL)

10%

Of Total CO2 Emission from Steel Sector

CBAM "Carbon Tax" by EU on Steel Imports

- Introduced in May 2023, applied on steel imports having higher carbon foot-print than EU-ETS Benchmark
- Will make **Indian Steel exports uncompetitive**, resulting in a profit decrease of $60-165/MT between 2026-2034

Metallurgical coke accounts for ~20% of total cost of Steel Making

- Import dependency
- Supply-Chain disruption
- Price fluctuation

TARGETS

Net Zero	42% reduction	50%
by 2070	by 2030	by 2047
Ministry of Steel, India	Emission target JSW Group	India's Scrap usage Target

ENABLERS

- Resource efficiency
- Green Hydrogen
- Scrap Usage
- Extensive Monitoring

Our Digital Solution: Sustainability Dashboard to track emissions across value chain with integrated Carbon offsetting, Scrap monitoring & CCUS monitoring

Raw Material (Iron Ore, Coking Coal, etc.) → Truck, Rail, Pipeline, Ship → **INTEGRATED STEEL PLANT** → Truck, Rail, Ship → **Customers** (End Users, Dealers etc.)

SUSTAINABILITY DASHBOARD

Scrap Usage Internal + External

Carbon Offsetting & Capture Internal + External

EMISSIONS MONITORING ACROSS THE VALUE-CHAIN

Scope 1 : Direct emissions from sources owned or controlled by JSW – Mines, BF Operations, Mills etc.	**Scope 2 :** Emissions from the production of energy source used by JSW for its operations	**Scope 3 :** Emissions from the activities that JSW is indirectly responsible for, up and down its value-chain

Carbon Offsetting & Capture (Internal + External)

- Track & Manage internal **carbon offsetting and capture** processes (Tree Planting + CCUS)
- Onboard external Carbon Positive NGOs for offsetting CO2 emissions through **Afforestation and Investment in climate action projects.**
- Onboard other Carbon Negative and Positive firms on the dashboard to create a **market-place for Carbon Credits**

Scrap Usage (Internal + External)

- Extensively **track Scrap/Waste generated** from processes across the value chain (CRM, HRM, Scrap produced at customers'(factories) end)
- Onboard **3rd party Scrap aggregators** to source scrap from landfills & Users/Factories
- Track and Manage Scrap Usage to **increase its percentage in steel making** process

Gen-AI Based Recommendations
To improve sustainable practices across the value-chain

Improved Visibility & Reporting
For intervention by top-management and compliance requirements

Sustainability Dashboard: Create data driven tool leveraging IoT, Cloud and GenAI for comprehensive tracking and reporting, unlocking value for all stakeholders

VALUE PROPOSITIONS

Standardise & Automate
emission data collection

Data Unification
across plants & geographies

Simplified
Internal tracking & reporting

Quick
compliance tracking & data accessibility

Diagnostic & Predictive Alerts
monitoring emissions

Easy
reporting for government approvals

SUSTAINABILITY DASHBOARD

MEASURE, MANAGE and REPORT

Navigation menu

- Choose from range of options to **navigate the dashboard**
- Collates important insights including option to view **scrap generation data, emissions tracker and important alerts**

Emission alerts

- Generate **alerts** when emissions increase above target level
- Provide **recommendations using Gen AI** to reduce emissions

Customization options

Choose from required **Granularity of detail**, plant **location** and **period**

Emissions overview

- Display **snapshots of emissions** during the selected period
- Select on a tile to view statistics in detail
- Generate **alerts** for emissions crossing the target level

Reports generation

Generate summary, pollution reports for **State Pollution Control Board**, and **sustainability reports**

IMPLEMENTATION

Automated data collection & consolidation

IOT Sensors

Edge Computing

Data Analysis

Cloud based analytics software

ML to identify patterns

Data visualization on Sustainability Dashboard

Generate emission metrics

Track emission, CCUS, carbon offset, scrap

VALUE FOR CUSTOMERS

Scan QR code from product packaging & JSW Banners

Get redirected to JSW website

View **sustainability report** made by dashboard

BENEFITS

- Increases **transparency**
- Enhances **customer engagement** & **brand equity**
- Exhibits **corporate responsibility**

Green Hydrogen in Steel Making: Implement green hydrogen powered DRI + EAF solution leveraging India's technological expertise in renewable energy to produce cost effective sustainable steel

Steel Making Processes in India (% Share)

BF-BOF	•Metallurgical coke used as reducing agent	**43%**
DRI - EAF/IF	• Non-coking coal, Natural Gas, SynGas or Coke-oven gas used as reducing agent •Used in Combination with EAF/IF	**32%**
EAF (Scrap Based)	•Electricity used to melt the scrap/recycled Iron	**25%**

- **H2** can be used to replace Natural Gas, SynGas and CO gas as reducing agent
- Electricity for EAF can be generate using **H2 fuel cell.**

India is at the Forefront of Renewable Energy Generation

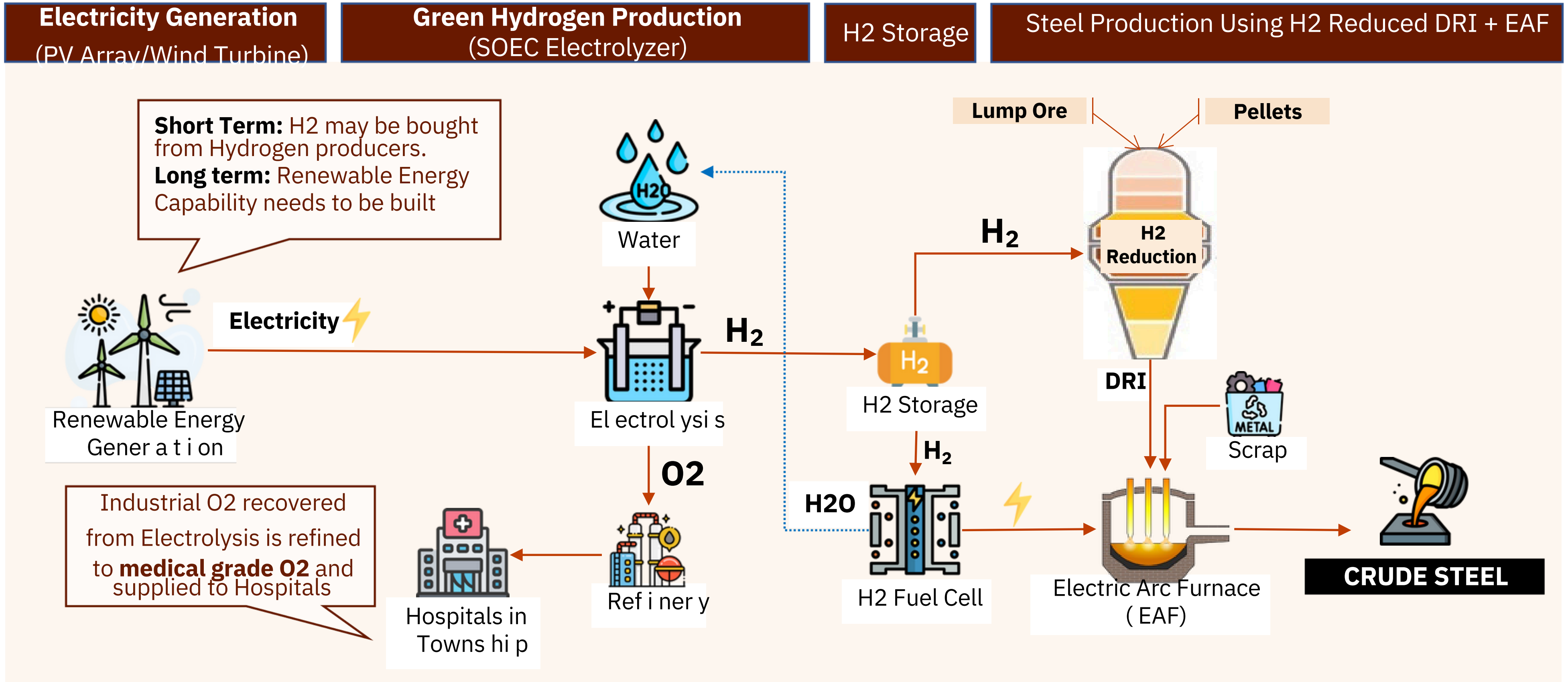
4th
Globally
in total renewable power capacity

$0.037/kWh
Levelized Cost of Electricity (LCOE)
of Renewable Energy (Solar)
(Lowest In the World)

179 GW
Renewable Capacity
(Fastest Growing in the World)

India's technological expertise in Renewable Energy Generation with the lowest levelized costs, can be leveraged to utilize Renewable power in Green H2 Generation in a cost effective and efficient manner

GREEN HYDROGEN POWERED STEEL MAKING



PROCESS HIGHLIGHTS

Renewable Energy
To Generate Green Hydrogen

Recycled Water
From Fuel-Cell & Waste-water treatment

Medical Oxygen
As a By-product of H_2 generation

Green Hydrogen future potential: Partner with technology enablers to decarbonize JSW's current 27.7 MTPA crude steel capacity through a total solar power installation of 16.7 GW with a phase-wise implementation

KEY TECHNOLOGY ENABLERS

Solar/Wind Energy Equip. Providers
SOLAR: SAATVIK, adani Renewables
WIND: SIEMENS Gamesa, SUZLON

Hydrogen Suppliers
Reliance Industries Limited, एनटीपीसी NTPC, HERO FUTURE ENERGIES, गेल GAIL

Electrolyser Provider
ohmium – India's 1st GreenHydrogen Electrolyser Gigafactory
AVAADA, Cummins

Potential Steel Production Using Green Hydrogen Route (Solar Power)

Assumptions

1. It takes **40—44 kWh** of Electricity to Produce 1 Kg of H2 though Electrolysis via SOEC Electrolyzer
2. **50 Kg of Green Hydrogen** is required to produce 1 Ton of Crude Steel
3. Average **10 Hrs. of daily output** from solar power plant, **3650 Hrs. of output in a Year**
4. SOEC Electrolyzer with **80% Efficiency**

Process (For a 10 GW Solar Plant)

10 GW Solar Plant → To Generate → **36500 GWh** Electricity Annually → To Produce → **0.83 MT** Hydrogen Annually → To Make → **16.6 MT** Crude Steel Annually

To completely **De-Carbonize** JSW's Operations: **27.7 MTPA** Crude Steel Capacity **Will Require** **16.7 GW** Solar Power Capacity

Building such Solar Capacity would require phase-wise implementation across 5-6 Years

POTENTIALS OF H2 IN STEEL MAKING

Lower Emissions
Study by McKinsey found that H2 could **reduce GHG emissions by up to 95%** from the steel industry

Energy Efficiency
Acc. to the IEA, direct reduction using H2 can **reduce energy consumption by ~20%** compared to traditional BF

Residue O2
The O2 captured as residue from Electrolysis can be refined to **produce medical grade O2** for hospitals

High-Quality Steel
Direct reduction produces high-quality iron that is **purer and has lower levels of impurities**, resulting in higher-quality steel.

LIMITATIONS OF H2 IN STEEL MAKING

High Input Costs
Using **H_2** to produce Steel is currently much more costly in India than using traditional methods

Infra. Challenges
Transitioning to H2-based steel requires extensive **modification to plants** which can be costly & time-consuming.

Safe Handling
H_2 is **difficult to handle** in the plant due to its flammability. Strict Safety measures are required

Water Scarcity
Producing **1 Kg of H2, requires 9 Kg of H_2O.** Water Scarcity in the country can be a challenge for Green H_2 production

Making processes future ready: Integrate GenAI in steel making process across the value-chain to improve demand forecast, reduce process waste and support workers in maintenance activities

Demand Forecasting & Inventory Management

Analyze past sales, market trends, inventory & supplier data for predictive modelling & accurate inventory levels

ISSUES

- Inaccurate forecasting models
- High carrying cost & obsolete inventory

SOLUTION

Data mining to identify patterns & relations ❯ **GenAI** ❯ **Predictive modelling** for optimal inventory

IMPACT

- Optimal inventory levels
- Waste reduction
- Cost Savings

Process Improvement & Quality control

Use GenAI to optimize machine input settings integrating with HSM II to detect & analyze root cause occurrences to suggest process improvements

ISSUES

- Time-taking defect analysis
- Redundant machine inputs

PROCESS FLOW

1. Analyze existing machine inputs
2. Track & log defects, analyze root cause
3. Change machine inputs, reducing defects
4. Improvement in process efficiency

IMPACT

- Improved quality control
- Enhanced efficiency
- Predictive Maintenance
- Cost Savings

"KaushalGyan" – GenAI based A ssistant

GenAI-driven app that improves staff experience by providing information at their fingertips

SOLUTION

GenAI:
- Industry best practices
- Knowledge management
- 24 X 7 support to staff
- Innovations across industry

IMPACT

- Data-driven Decision making
- Improved safety
- Enhanced User training

USE CASE

USS MineMind GenAI

First gen AI-driven application powered by Google AI that aims to simplify equipment maintenance by providing optimal solutions for mechanical problems, saving time and money, and ultimately improving productivity

Technology adoption: Implement strategies across key operational levers to drive technology integration with focus on alleviating negative impact on employees with **cross training, sequential implementation and On-ground assistance**

TRANSITION FRAMEWORK FOR TECHNOLOGY INTRODUCTION

Technology Introduction

Impact on employees

Skills gap + Fear of job loss + Loss of autonomy + Resistance to change

Key intervention areas

Social Factors

Physical S etting

Technology

Organizational Arrangements

Result of intervention

CHANGE IN BEHAVIOR

INCREASED ACCEPTANCE OF TECHNOLOGY

Organizational & Individual improvement

Organizational Performance ↔ Individual Development

STRATEGIES FOR INTERVENTION

SOCIAL FACTORS

1 Cross training

Identify employees who might lose jobs due to technology → **Cross train** them for other jobs with better person-job fit

Benefits: maintain employee trust, enhance adaptability

PHYSICAL SETTING

2 Sequential implementation

Launch technology for **single plant** and optimize the strategy → Next launch for other sites and implement **best practices** learnt from pilot plant

Benefits: identify & address challenges at small scale, smooth transition

TECHNOLOGY

3 Increase transparency around digitalization

Explain the **working behind technology** to employees during training → Collect and incorporate **employee feedback** for technology integration

Benefits: alleviate employee concerns, infuse sense of ownership

ORGANIZATIONAL ARRANGEMENTS

4 On-ground assistance

Create on-ground team- ***"Saarthi team"*** with technology proficient workers → Team will assist employees at **on job challenges for initial 3 months** post training

Benefits: boost employee confidence, reduce frustration

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Thank You

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